

BOUNDED RATIOS OF LORENTZIAN POLYNOMIALS II: THE COMPLETE QUADRATIC LOCAL-TO-GLOBAL CLASSIFICATION

DIJIA CHEN, BOWEN GAN, IVY LIU, ZEMENG WANG, AND CHENGZHI WU

ABSTRACT. Every quadratic Hessian slice of a Lorentzian polynomial yields bounded monomial ratios among the normalized coefficients of the polynomial. We determine exactly for which pairs (n, d) these quadratic-slice ratios generate the full bounded-ratio cone for every M -convex support $S \subseteq \Delta_n^d$. For $d \geq 2$, this quadratic local-to-global principle holds universally if and only if

$$n \leq 3, \quad d = 2, \quad \text{or} \quad (n, d) = (4, 3).$$

In every remaining case, the principle fails already for Lorentzian polynomials with full support: for cubics in $n \geq 5$ variables and for polynomials of degree $d \geq 4$ in $n \geq 4$ variables. We identify the two minimal obstructions, at $(n, d) = (4, 4)$ and $(n, d) = (5, 3)$, and propagate them using cut-cone certificates and variable-lifting arguments. The quartic bounded ratio extends to every higher degree by differentiation, while perturbing a square-transportation direction yields an explicit degree-uniform family of separating functionals. As a conceptual byproduct, we show that transportation costs on finite median graphs give polar directions in arbitrary dimension and degree.

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1. INTRODUCTION

Lorentzian polynomials, introduced by Brändén and Huh [4], provide a common framework for Hodge–Riemann-type inequalities in combinatorics and convex geometry. A homogeneous polynomial of degree $d \geq 2$ with nonnegative coefficients is Lorentzian if and only if its support is M -convex and every quadratic derivative has Hessian with at most one positive eigenvalue [4, Theorem 2.25]. Thus, once the support is fixed, Lorentzianity is determined by a collection of quadratic Hessian slices.

This local characterization raises a natural question for multiplicative inequalities among coefficients. Let

$$f(\mathbf{x}) = \sum_{\alpha \in S} c_{\alpha} \frac{\mathbf{x}^{\alpha}}{\alpha!}, \quad c_{\alpha} > 0,$$

and, for $w = (w_{\alpha})_{\alpha \in S} \in \mathbb{R}^S$, write

$$R_w(f) := \prod_{\alpha \in S} c_{\alpha}^{w_{\alpha}}.$$

We call w a *bounded ratio* if R_w is uniformly bounded on the Lorentzian locus $\mathcal{L}_S^+$. Every quadratic Hessian slice has its own bounded-ratio cone, and extending these local exponent vectors to the global coefficient space gives a cone

$$\text{Cone}(\mathcal{G}_n(S)) \subseteq \text{BR}(\mathcal{L}_S^+).$$

We ask when the reverse inclusion holds: when is every global bounded ratio generated by bounded ratios already visible on the quadratic slices?

Part I [5] answers this question in three variables. It proves that

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}_3(S))$$

for every degree and every M -convex support $S \subseteq \Delta_3^d$. The purpose of the present paper is to determine exactly how far this quadratic local-to-global principle extends beyond the ternary setting.

Our main result gives a complete classification.

Theorem 1.1 (Complete quadratic local-to-global classification). *Let $n \geq 1$ and $d \geq 2$. The following are equivalent.*

(i) *For every nonempty M -convex support $S \subseteq \Delta_n^d$,*

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}_n(S)).$$

(ii) *One has*

$$n \leq 3, \quad \text{or} \quad d = 2, \quad \text{or} \quad (n, d) = (4, 3).$$

Moreover, whenever (ii) fails, the local-to-global principle already fails for full support:

$$\text{Cone}(\mathcal{G}_n(\Delta_n^d)) \subsetneq \text{BR}(\mathcal{L}_{n,d}^+).$$

Thus the only nonquadratic positive case beyond three variables is the quaternary cubic case. Equivalently, the classification may be displayed as

	$d = 2$	$d = 3$	$d \geq 4$
$n \leq 3$	✓	✓	✓
$n = 4$	✓	✓	×
$n \geq 5$	✓	×	×

where ✓ means equality for every M -convex support and × means failure already for full support.

The proof has one exceptional positive case and two minimal negative cases. The positive case is $(n, d) = (4, 3)$. Unlike the ternary case, a quadratic slice in four variables can exhibit a genuine four-point branching phenomenon. We analyze the possible M -convex cubic supports and reduce the branching case to a finite support-graph problem. On the dual side, the relevant local polar directions admit a nonnegative decomposition into M -concave directions. Exponential degeneration along such a direction then rules out any bounded ratio outside the quadratic local cone. This proves the local-to-global theorem for every M -convex support in Δ_4^3 .

The failure region has, with respect to the coordinatewise order on pairs (n, d) , exactly two minimal elements:

$$(4, 4) \quad \text{and} \quad (5, 3).$$

For each of these pairs we construct an explicit full-support bounded ratio which does not belong to the quadratic local cone. The two constructions have a common structure. Boundedness follows from quadratic matrix inequalities, while nonlocality is certified by a separating functional whose restrictions to the quadratic slices are controlled by cut semimetrics. This leads to a convenient cut-cone criterion for recognizing elements of the polar of the quadratic local cone.

We then propagate the four-variable quartic obstruction through every higher degree by differentiation together with a degree-uniform family of separating functionals. A variable-aggregation lemma propagates each obstruction with a cut-cone certificate to every larger number of variables. Starting from $(4, 4)$ and $(5, 3)$, these arguments produce full-support counterexamples throughout the entire failure region. Together with the positive cases, this proves Theorem 1.1.

Huang, Huh, Soskin, and Wang [6] study bounded ratios for Lorentzian matrices and their relation to the cut cone. Their four-point inequalities motivate the cut-semimetric certificates used in our obstruction arguments. Baldi and Kummer [2] identify, for a fixed M -convex support, the bounded-ratio cone of Lorentzian polynomials with the dual cone of M -convex functions. From this perspective, Theorem 1.1 determines exactly when this global cone is generated by the bounded-ratio cones of the quadratic Hessian slices.

The paper is organized as follows. Section 2 fixes notation, defines the quadratic local cone, and settles the easy positive cases. Section 3 proves the exceptional positive case $(n, d) = (4, 3)$ for arbitrary M -convex support. Section 4 develops the cut-cone separation certificates used for the counterexamples. Section 5 constructs the two minimal full-support obstructions at $(4, 4)$ and $(5, 3)$. Section 6 first propagates in degree, then propagates in the number of variables, and finally completes the proof of Theorem 1.1. Section 7 gives a median-graph interpretation of the transportation directions underlying the separating functionals.

2. PRELIMINARIES AND THE EASY POSITIVE CASES

Throughout, $\mathbb{N} := \mathbb{Z}_{\geq 0}$. For a finite set S , let ε_α denote the standard basis vector of $\mathbb{R}^S$ indexed by $\alpha \in S$, and use the pairing

$$\langle w, q \rangle = \sum_{\alpha \in S} w_\alpha q_\alpha.$$

For $C \subseteq \mathbb{R}^S$, our polar convention is

$$C^\circ := \{q \in \mathbb{R}^S : \langle w, q \rangle \leq 0 \text{ for all } w \in C\}.$$

We write $\text{Cone}(E)$ for the set of finite nonnegative linear combinations of elements of E . For $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, set

$$|\alpha| = \alpha_1 + \dots + \alpha_n, \quad [n] = \{1, \dots, n\}, \quad \Delta_n^d = \{\alpha \in \mathbb{N}^n : |\alpha| = d\}.$$

Let e_i denote the i -th standard basis vector of $\mathbb{R}^n$. For $S \subseteq \Delta_n^d$, we use factorial normalization

$$f(\mathbf{x}) = \sum_{\alpha \in S} c_\alpha \frac{\mathbf{x}^\alpha}{\alpha!}, \quad \alpha! = \alpha_1! \cdots \alpha_n!,$$

so that the Hessian of each quadratic derivative is read directly from the normalized coefficients.

Definition 2.1 (M -convex support). A nonempty set $S \subseteq \Delta_n^d$ is M -convex if, whenever $\alpha, \beta \in S$ and $\alpha_i > \beta_i$, there is an index j with $\alpha_j < \beta_j$ such that

$$\alpha - e_i + e_j \in S, \quad \beta - e_j + e_i \in S.$$

For $d \geq 2$ and $\beta \in \Delta_n^{d-2}$, define the quadratic slice support

$$S_\beta := \{e_i + e_j : i, j \in [n], \beta + e_i + e_j \in S\} \subseteq \Delta_n^2$$

and the corresponding symmetric Hessian array

$$H_\beta(f) = (c_{\beta+e_i+e_j})_{1 \leq i, j \leq n},$$

with the convention that $c_\gamma = 0$ outside S . Every nonempty S_β is M -convex: applying the exchange axiom in S to $\beta + \gamma$ and $\beta + \eta$, and then subtracting β , gives the required exchange in S_β . By Brändén–Huh [4, Theorem 2.25], f is Lorentzian if and only if S is M -convex and every $H_\beta(f)$ has at most one positive eigenvalue.

All theorem numbers cited from [4] refer to the revised version arXiv:1902.03719v8; the bibliography also lists the published version.

For a fixed support S , let $\mathcal{L}_S^+$ be the class of Lorentzian polynomials whose normalized coefficients are positive exactly on S . For $w \in \mathbb{R}^S$, define

$$R_w(f) := \prod_{\alpha \in S} c_\alpha^{w_\alpha}, \quad \text{BR}(\mathcal{L}_S^+) := \left\{ w \in \mathbb{R}^S : \sup_{f \in \mathcal{L}_S^+} R_w(f) < \infty \right\}.$$

It follows directly from the definition that $\text{BR}(\mathcal{L}_S^+)$ is a convex cone. For full support, we abbreviate

$$\mathcal{L}_{n,d}^+ := \mathcal{L}_{\Delta_n^d}^+.$$

If

$$v = \sum_{\gamma \in S_\beta} v_\gamma \varepsilon_\gamma \in \mathbb{R}^{S_\beta},$$

write

$$\tilde{v}^\beta := \sum_{\gamma \in S_\beta} v_\gamma \varepsilon_{\beta+\gamma} \in \mathbb{R}^S.$$

The *quadratic local generators* are

$$\mathcal{G}_n(S) := \bigcup_{\substack{\beta \in \Delta_n^{d-2} \\ S_\beta \neq \emptyset}} \left\{ \tilde{v}^\beta : v \in \text{BR}(\mathcal{L}_{S_\beta}^+) \right\}.$$

Thus, $\text{Cone}(\mathcal{G}_n(S))$ is the cone generated by all bounded ratios coming from quadratic Hessian slices. Since every Hessian slice of a Lorentzian polynomial is Lorentzian,

$$(2.1) \quad \text{Cone}(\mathcal{G}_n(S)) \subseteq \text{BR}(\mathcal{L}_S^+)$$

for every M -convex support S .

We shall also use M -concave functions. A function $q : S \rightarrow \mathbb{R}$ on an M -convex set is M -concave if, whenever $\alpha_i > \beta_i$, one can choose j as in the exchange axiom so that

$$q(\alpha) + q(\beta) \leq q(\alpha - e_i + e_j) + q(\beta + e_i - e_j).$$

It is enough to check the local exchanges with $\|\alpha - \beta\|_1 = 4$ [7, Theorem 6.4]. We use the standard degeneration theorem [4, Theorem 3.14]: if q is M -concave, then

$$(2.2) \quad f_t(\mathbf{x}) = \sum_{\alpha \in S} e^{tq(\alpha)} \frac{\mathbf{x}^\alpha}{\alpha!}$$

belongs to $\mathcal{L}_S^+$ for every $t \geq 0$. In the notation of [4, Theorem 3.14], this is the choice $\nu = -q$ and $u = e^{-t} \in (0, 1]$, so that $u^{\nu(\alpha)} = e^{tq(\alpha)}$.

Proposition 2.2 (The easy positive cases). *The quadratic local-to-global equality holds for every M -convex support in each of the following cases:*

- (i) $d = 2$ and arbitrary n ;
- (ii) $n \leq 3$ and arbitrary $d \geq 2$.

Proof. If $d = 2$, the only slice is $\beta = 0$, so the definition of $\mathcal{G}_n(S)$ gives

$$\text{Cone}(\mathcal{G}_n(S)) = \text{BR}(\mathcal{L}_S^+).$$

For $n = 3$, the assertion is the ternary local-to-global theorem of [5, Theorem 3.6]. The cases $n = 1, 2$ follow by adjoining zero variables: identifying $S \subseteq \Delta_n^d$ with the corresponding coordinate face of Δ_3^d preserves the coefficient locus, Lorentzianity of all quadratic slices, and the local cone. $\square$

3. THE EXCEPTIONAL POSITIVE CASE: QUATERNARY CUBICS

The only nonquadratic positive case with more than three variables is $(n, d) = (4, 3)$. The proof is more delicate because a quadratic slice can contain all six off-diagonal points, producing a genuine four-point branching phenomenon.

Theorem 3.1 (Quaternary cubics). *For every nonempty M -convex support $S \subseteq \Delta_4^3$,*

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}_4(S)).$$

The proof is based on separation. We first extract a finite family of elementary bounded-ratio directions from the quadratic slices. If no slice is branching, every functional that is nonpositive on these directions is M -concave. If a branching slice occurs, support reduction either reduces the problem to the quadratic case or puts the support in a form that can be encoded by a directed graph. In the latter case, the star decomposition expresses the relevant functional as a finite nonnegative combination of M -concave functions. Exponential degeneration then rules out a separating functional for a global bounded ratio.

3.1. Elementary local directions and support reduction. We first record the quadratic inequalities used throughout the paper. In this section, they provide the elementary local directions needed for the quaternary cubic argument; they will also supply the coefficient bounds for the two minimal obstructions. The four-point inequality appears in [6, Proposition 3.1]. The assertions of the following lemma also appear, in an equivalent formulation, in Baker–Huh–Kummer–Lorscheid [1, Lemma 5.15]. We include a direct proof for completeness.

Lemma 3.2 (Quadratic inequalities). *Let $n \geq 1$, and let $P = (p_{ij})_{1 \leq i, j \leq n}$ be symmetric and entrywise nonnegative, with at most one positive eigenvalue. For distinct i, j ,*

$$p_{ii}p_{jj} \leq p_{ij}^2,$$

and for pairwise distinct i, j, k ,

$$p_{ii}p_{jk} \leq 2p_{ij}p_{ik}.$$

If i, j, k, ℓ are pairwise distinct, then

$$\sqrt{p_{ij}p_{k\ell}}, \quad \sqrt{p_{ik}p_{j\ell}}, \quad \sqrt{p_{i\ell}p_{jk}}$$

are the side lengths of a possibly degenerate triangle. In particular, if one of the three opposite-edge products vanishes, then the other two are equal.

Proof. The principal 2×2 submatrix on $\{i, j\}$ has at most one positive eigenvalue, and hence has nonpositive determinant. This gives $p_{ii}p_{jj} \leq p_{ij}^2$.

For the second inequality, decrease the j - and k -diagonal entries of the principal submatrix on $\{i, j, k\}$ to zero. Subtracting a positive semidefinite matrix cannot increase the number of positive eigenvalues, so the resulting matrix still has at most one positive eigenvalue. If $p_{jk} > 0$, its determinant is nonnegative, while

$$\det \begin{pmatrix} p_{ii} & p_{ij} & p_{ik} \\ p_{ij} & 0 & p_{jk} \\ p_{ik} & p_{jk} & 0 \end{pmatrix} = p_{jk}(2p_{ij}p_{ik} - p_{ii}p_{jk}).$$

The case $p_{jk} = 0$ is immediate.

For the four-point assertion, pass to the 4×4 principal submatrix on $\{i, j, k, \ell\}$, relabel its indices as $1, 2, 3, 4$, and denote this submatrix again by P . Set

$$B = P - \text{diag}(p_{11}, p_{22}, p_{33}, p_{44}).$$

Again, B has at most one positive eigenvalue. Since B has zero diagonal and hence trace zero, it follows that $\det B \leq 0$. Set

$$A = p_{12}p_{34}, \quad C = p_{13}p_{24}, \quad E = p_{14}p_{23}.$$

Expanding the determinant gives

$$\det B = A^2 + C^2 + E^2 - 2AC - 2AE - 2CE = (A - C - E)^2 - 4CE.$$

Consequently,

$$(\sqrt{C} - \sqrt{E})^2 \leq A \leq (\sqrt{C} + \sqrt{E})^2,$$

and therefore

$$|\sqrt{C} - \sqrt{E}| \leq \sqrt{A} \leq \sqrt{C} + \sqrt{E}.$$

These are precisely the triangle inequalities, including the degenerate case. If one side length vanishes, the other two coincide. $\square$

For a nonempty M -convex support $T \subseteq \Delta_4^2$, let $\mathcal{E}(T) \subseteq \mathbb{R}^T$ consist of all vectors of the forms

$$(3.1) \quad \varepsilon_{2e_i} + \varepsilon_{2e_j} - 2\varepsilon_{e_i+e_j},$$

$$(3.2) \quad \varepsilon_{2e_i} + \varepsilon_{e_j+e_k} - \varepsilon_{e_i+e_j} - \varepsilon_{e_i+e_k},$$

whenever all displayed points belong to T , where $i \neq j$ in (3.1) and i, j, k are pairwise distinct in (3.2). We also include both signs of

$$(3.3) \quad \varepsilon_{e_i+e_j} + \varepsilon_{e_k+e_\ell} - \varepsilon_{e_i+e_k} - \varepsilon_{e_j+e_\ell}$$

whenever i, j, k, ℓ are pairwise distinct, the four displayed points belong to T , and at least one of $e_i + e_\ell$ and $e_j + e_k$ does not belong to T .

Lemma 3.2 shows that every vector in $\mathcal{E}(T)$ is a bounded-ratio direction on $\mathcal{L}_T^+$. The bounds associated with (3.1) and (3.2) are 1 and 2, respectively, while (3.3) is an equality direction.

From now on, fix a nonempty M -convex support $S \subseteq \Delta_4^3$. Write $S_i := S_{e_i}$ and define

$$\mathcal{G}_4^{\text{el}}(S) := \{\tilde{g}^{e_i} : i \in [4], S_i \neq \emptyset, g \in \mathcal{E}(S_i)\}.$$

Every $h \in \mathcal{G}_4^{\text{el}}(S)$ is balanced, in the sense that

$$(3.4) \quad \sum_{\alpha \in S} h_{\alpha} \alpha = 0.$$

Moreover,

$$(3.5) \quad \text{Cone}(\mathcal{G}_4^{\text{el}}(S)) \subseteq \text{Cone}(\mathcal{G}_4(S)) \subseteq \text{BR}(\mathcal{L}_S^+).$$

The first inclusion follows from the definition of $\mathcal{E}(S_i)$; the second follows because every quadratic Hessian slice of a Lorentzian cubic is Lorentzian.

Call a nonempty slice S_i *branching* if all six off-diagonal points $e_j + e_k$, $1 \leq j < k \leq 4$, belong to S_i . Put

$$\mathbf{1} := (1, 1, 1, 1), \quad z_i := \mathbf{1} - e_i.$$

Lemma 3.3 (Support reduction). *Suppose that S_i is branching. Then either $S = e_i + T$ for some nonempty M -convex set $T \subseteq \Delta_4^2$, or*

$$z_1, z_2, z_3, z_4 \in S.$$

Proof. Since S_i is branching, $z_j \in S$ for every $j \neq i$. If $\alpha_i \geq 1$ for every $\alpha \in S$, then

$$S = e_i + T, \quad T := S - e_i \subseteq \Delta_4^2,$$

and translation by e_i shows directly from the symmetric exchange axiom that T is M -convex.

Otherwise, choose $\alpha \in S$ with $\alpha_i = 0$. If $\alpha = z_i$, there is nothing to prove. Since $|\alpha| = 3$, some $j \neq i$ satisfies $\alpha_j \geq 2$. Suppose first that $\alpha = 2e_j + e_k$, and let ℓ be the remaining index. Compare α with $z_j = e_i + e_k + e_{\ell}$. At the j -coordinate, the exchange axiom can use only i or ℓ . In the first case,

$$z_j - e_i + e_j = z_i \in S,$$

whereas in the second case,

$$\alpha - e_j + e_{\ell} = z_i \in S.$$

If $\alpha = 3e_j$, compare α with z_j . If the exchange uses i , then again $z_j - e_i + e_j = z_i \in S$. Otherwise it produces $2e_j + e_k \in S$ for some $k \notin \{i, j\}$, and the preceding case applies. $\square$

Lemma 3.4 (Translated quadratic supports). *Suppose that $S = e_i + T$, where $T \subseteq \Delta_4^2$ is nonempty and M -convex. Differentiation with respect to x_i preserves the normalized coefficient array and*

$$f \in \mathcal{L}_S^+ \iff \partial_i f \in \mathcal{L}_T^+.$$

Consequently, under the natural identification of coefficient coordinates,

$$\text{BR}(\mathcal{L}_S^+) = \text{BR}(\mathcal{L}_T^+).$$

Proof. In factorial normalization,

$$\partial_i \left(\sum_{\gamma \in T} c_{e_i + \gamma} \frac{\mathbf{x}^{e_i + \gamma}}{(e_i + \gamma)!} \right) = \sum_{\gamma \in T} c_{e_i + \gamma} \frac{\mathbf{x}^{\gamma}}{\gamma!}.$$

Thus the two normalized coefficient arrays agree. The support condition is also preserved, since $S = e_i + T$. If $j \neq i$, every nonzero quadratic derivative $\partial_j f$ is divisible by x_i ; after placing the i th coordinate first, its Hessian has the form

$$\begin{pmatrix} a & b^{\top} \\ b & 0 \end{pmatrix}.$$

Interlacing with the zero 3×3 principal submatrix shows that this Hessian has at most one positive eigenvalue. The Hessian of $\partial_i f$ is the normalized coefficient matrix of $\partial_i f$. The characterization in [4, Theorem 2.25] therefore gives the stated equivalence. $\square$

3.2. Branching supports and the star decomposition. Assume for the rest of this part that $z_1, z_2, z_3, z_4 \in S$. For $i \neq j$, set

$$a_{ij} := 2e_i + e_j, \quad p_i := 3e_i, \quad N_i := \{j \neq i : a_{ij} \in S\}.$$

Together with the z_i , the points a_{ij} and p_i exhaust Δ_4^3 . We write $i \rightarrow j$ when $a_{ij} \in S$, and $i \leftrightarrow j$ when both $i \rightarrow j$ and $j \rightarrow i$ occur.

Lemma 3.5 (Support graph). *For every $i \in [4]$,*

$$|N_i| \in \{0, 2, 3\}, \quad p_i \in S \implies |N_i| = 3.$$

Proof. Suppose that $a_{ij} \in S$, and let k, ℓ be the remaining indices. Compare a_{ij} with $z_j = e_i + e_k + e_\ell$. Symmetric exchange at the j -coordinate forces $a_{ik} \in S$ or $a_{i\ell} \in S$. Hence $|N_i| \neq 1$.

Now suppose that $p_i \in S$. Fix $j \neq i$, and let k, ℓ be the remaining indices. Symmetric exchange between p_i and z_j at the i -coordinate forces both a_{ik} and $a_{i\ell}$ to lie in S . Varying j gives $|N_i| = 3$. $\square$

Let $q = (q_\alpha)_{\alpha \in S} \in \mathbb{R}^S$. The vectors $z_1, \dots, z_4$ form a basis of $\mathbb{R}^4$, since the matrix with these vectors as columns is $J - I$. Hence there is a unique linear function $\ell(\alpha) = \langle b, \alpha \rangle$ such that

$$(q - \ell)(z_i) = 0 \quad \text{for every } i \in [4].$$

For such a normalized function q , define

$$s_{ij} := -q(a_{ij}) \quad \text{when } a_{ij} \in S, \quad r_i := -q(p_i) \quad \text{when } p_i \in S.$$

Lemma 3.6 (Local exchange criterion). *Assume that $q(z_i) = 0$ for every $i \in [4]$. If*

$$q \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ,$$

then, whenever the displayed coordinates are defined,

$$(3.6) \quad s_{ij} \geq 0,$$

$$(3.7) \quad s_{ij} \leq s_{ik} + s_{ji},$$

$$(3.8) \quad s_{ij} + s_{ik} \leq r_i.$$

Here $i \neq j$ in (3.6), and i, j, k are pairwise distinct in (3.7) and (3.8). Each inequality is imposed only when all of its coefficient coordinates belong to S . If $N_i = \{j, k\}$, then

$$(3.9) \quad s_{ij} = s_{ik}.$$

Moreover, among normalized functions satisfying (3.6)–(3.9), M -concavity is equivalent to the additional condition

$$(3.10) \quad \min_{j \in N_i} s_{ij} \text{ is attained at least twice whenever } |N_i| = 3.$$

Proof. Fix $m \in [4]$ and write

$$Q_m(u, v) := q(e_m + e_u + e_v).$$

For distinct $j, k \neq m$,

$$Q_m(m, m) = -r_m, \quad Q_m(m, j) = -s_{mj}, \quad Q_m(j, j) = -s_{jm}, \quad Q_m(j, k) = 0.$$

Indeed, if ℓ is the remaining index, then $e_m + e_j + e_k = z_\ell$.

Let j, k, ℓ be the three indices distinct from m . Pairing q with the elementary triangular directions determined, respectively, by

$$(2e_j, e_k + e_\ell), \quad (2e_j, e_m + e_k), \quad (2e_m, e_j + e_k)$$

gives (3.6), (3.7), and (3.8). Each pairing is used only when all of the corresponding coefficient coordinates are present. If $N_m = \{j, k\}$, then $e_m + e_\ell \notin S_m$, whereas the four points defining the corresponding rectangle belong to S_m . The degenerate four-point direction (3.3), included with both signs, therefore gives $s_{mj} = s_{mk}$. This proves (3.9).

It remains to identify the local M -concavity conditions. Every local exchange in Δ_4^3 lies in a quadratic slice. Indeed, if $\alpha, \beta \in \Delta_4^3$ and $\|\alpha - \beta\|_1 = 4$, then

$$|\alpha \wedge \beta| = \frac{|\alpha| + |\beta| - \|\alpha - \beta\|_1}{2} = 1,$$

so $\alpha \wedge \beta = e_m$ for some m . The diagonal–off-diagonal exchanges give (3.6)–(3.8). If $j, k \neq m$, the diagonal–diagonal exchange between $2e_j$ and $2e_k$ gives

$$-s_{jm} - s_{km} \leq 0,$$

which follows from (3.6). The only remaining diagonal–diagonal inequality is

$$2s_{mj} \leq r_m + s_{jm},$$

coming from $2e_m$ and $2e_j$. If this pair occurs, then $p_m \in S$, so $|N_m| = 3$ by Lemma 3.5. Choose $k \in N_m \setminus \{j\}$. Adding

$$r_m \geq s_{mj} + s_{mk} \quad \text{and} \quad s_{mj} \leq s_{mk} + s_{jm}$$

proves the required inequality.

Only exchanges between disjoint off-diagonal pairs remain. If $|N_m| = 2$, the unique feasible branch gives (3.9). If $|N_m| = 3$, the three opposite pairings have values $-s_{mj}, -s_{mk}, -s_{m\ell}$, so the exchange condition is precisely (3.10). This exhausts all local exchanges. Since it is enough to check local exchanges for M -concavity, the last assertion follows. $\square$

The preceding lemma isolates the only branching obstruction: a function in $\text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ$ need not itself satisfy (3.10). The next lemma removes this obstruction. Its atoms have four distinct roles: star atoms remove the common row minima, two-cycle atoms remove symmetric residuals on mutual pairs, one-edge atoms remove the remaining nonsymmetric residuals, and pure r_i atoms remove the final slack.

Lemma 3.7 (Star decomposition). *Let $q : S \rightarrow \mathbb{R}$ be normalized by $q(z_i) = 0$ for every i , and suppose that its coordinates satisfy (3.6)–(3.9). Then there are finitely many M -concave functions $q^{(\rho)} : S \rightarrow \mathbb{R}$ and coefficients $\lambda_\rho \geq 0$ such that*

$$q = \sum_{\rho} \lambda_\rho q^{(\rho)}.$$

Proof. For each i , define

$$m_i := \begin{cases} 0, & N_i = \emptyset, \\ s_{ij} \text{ (} = s_{ik} \text{)}, & N_i = \{j, k\}, \\ \min_{j \in N_i} s_{ij}, & |N_i| = 3, \end{cases} \quad u_{ij} := s_{ij} - m_i.$$

Then $u_{ij} \geq 0$, and

$$(3.11) \quad u_{ij} > 0 \implies |N_i| = 3.$$

Indeed, if $|N_i| = 2$, the two entries in row i are equal by (3.9).

For a mutual pair $i \leftrightarrow j$, set

$$e_{ij} := \min\{u_{ij}, u_{ji}\} = e_{ji}, \quad y_{ij} := (u_{ij} - u_{ji})_+.$$

For a non-mutual ordered pair, set $y_{ij} := 0$. The key estimate is

$$(3.12) \quad y_{ij} \leq m_j.$$

Indeed, if $y_{ij} > 0$, then $u_{ij} > 0$, so $|N_i| = 3$ by (3.11). Since j is not a minimizer in row i , there is some $k \neq i, j$ such that $u_{ik} = 0$. Using (3.7), we obtain

$$m_i + u_{ij} = s_{ij} \leq s_{ik} + s_{ji} = m_i + m_j + u_{ji},$$

which proves (3.12).

We now decompose the s -coordinates. Terms with zero coefficient are omitted throughout. Each atom below is a function on S with $q(z_i) = 0$; it is specified by listing its nonzero s - and r -coordinates, and all unspecified coordinates are zero.

For fixed j , let

$$I_j := \{i \neq j : i \leftrightarrow j, y_{ij} > 0\}.$$

If $m_j = 0$, then $I_j = \emptyset$ by (3.12), and there is no contribution centered at j . Assume that $m_j > 0$, and put

$$\theta_{ij} := \frac{y_{ij}}{m_j} \quad (i \in I_j).$$

For each $J \subseteq I_j$, set

$$\pi_{j,J} := \prod_{i \in J} \theta_{ij} \prod_{i \in I_j \setminus J} (1 - \theta_{ij}).$$

Then

$$\sum_{J \subseteq I_j} \pi_{j,J} = 1, \quad \sum_{\substack{J \subseteq I_j \\ i \in J}} \pi_{j,J} = \theta_{ij} \quad (i \in I_j).$$

Define the star atom $H_{j,J}$ by

$$s_{jk} = 1 \quad (k \in N_j), \quad s_{ij} = 1 \quad (i \in J),$$

together with

$$r_j = 2 \quad \text{if } p_j \in S, \quad r_i = 1 \quad \text{if } i \in J \text{ and } p_i \in S.$$

The sum

$$\sum_{J \subseteq I_j} m_j \pi_{j,J} H_{j,J}$$

contributes m_j to every outgoing edge from j and y_{ij} to every asymmetric incoming edge $i \rightarrow j$.

After subtracting all star atoms, a mutual pair $i \leftrightarrow j$ has the same residual e_{ij} in both directions. For each unordered mutual pair with $e_{ij} > 0$, include the term $e_{ij} C_{ij}$, where the two-cycle atom C_{ij} is defined by

$$s_{ij} = s_{ji} = 1, \quad r_i = r_j = 1 \quad \text{whenever the corresponding } p_i, p_j \text{ belong to } S.$$

If $i \rightarrow j$ is non-mutual and $u_{ij} > 0$, include the term $u_{ij} E_{ij}$, where the one-edge atom E_{ij} is defined by

$$s_{ij} = 1, \quad r_i = 1 \quad \text{if } p_i \in S.$$

At this point, all s -coordinates have been matched.

Suppose that $p_i \in S$. Then $|N_i| = 3$ and $\min_{j \in N_i} u_{ij} = 0$. Hence

$$\max_{\substack{j, k \in N_i \\ j \neq k}} (s_{ij} + s_{ik}) = 2m_i + \sum_{j \in N_i} u_{ij}.$$

By (3.8),

$$r_i \geq 2m_i + \sum_{j \in N_i} u_{ij}.$$

The atoms already chosen contribute exactly the right-hand side to the r_i -coordinate. Thus the remaining slack

$$\delta_i := r_i - 2m_i - \sum_{j \in N_i} u_{ij} \geq 0$$

is accounted for by δ_i times the pure r_i atom, whose only nonzero coordinate is $r_i = 1$. Since the points z_i , a_{ij} , and p_i exhaust S , this gives an exact finite nonnegative decomposition of q .

It remains to check that every atom occurring with positive coefficient is M -concave. If C_{ij} occurs, then $e_{ij} > 0$, and (3.11) gives $|N_i| = |N_j| = 3$. If E_{ij} occurs, then $u_{ij} > 0$, so $|N_i| = 3$. A leaf row of a star also has three entries, since $i \in I_j$ implies $u_{ij} > 0$. By Lemma 3.6, it is enough to verify (3.6)–(3.10). The nonzero rows of these atoms are of the forms

$$(1, 1), \quad (1, 1, 1), \quad (1, 0, 0).$$

Thus (3.6), (3.9), and (3.10) are immediate. The chosen r -coordinates give (3.8). For (3.7), whenever the left-hand side is 1, either another center edge or the reverse edge contributes 1 on the right-hand side. For a one-edge atom, the reverse edge is absent, so no such support-defined instance of (3.7) occurs. Pure r_i atoms also satisfy all five conditions. Therefore every selected atom is M -concave. $\square$

3.3. Proof of the quaternary cubic theorem.

Proof of Theorem 3.1. The second inclusion in (3.5) gives

$$\text{Cone}(\mathcal{G}_4(S)) \subseteq \text{BR}(\mathcal{L}_S^+).$$

Let $w = (w_\alpha)_{\alpha \in S} \in \text{BR}(\mathcal{L}_S^+)$. We first record the balance relation

$$(3.13) \quad \sum_{\alpha \in S} w_\alpha \alpha = 0.$$

Indeed, the choice $q \equiv 0$ in (2.2) shows that $\mathcal{L}_S^+$ is nonempty. For $a \in \mathbb{R}^4$ and $t \in \mathbb{R}$, the change of variables $x_i \mapsto e^{ta_i} x_i$ preserves $\mathcal{L}_S^+$ and multiplies R_w by

$$\exp \left(t \left\langle a, \sum_{\alpha \in S} w_\alpha \alpha \right\rangle \right).$$

Boundedness for all $t \in \mathbb{R}$ gives (3.13).

The set $\mathcal{G}_4^{\text{el}}(S)$ is finite. Hence its conic hull is closed, and strict separation applies.

Suppose first that no slice is branching. If

$$w \notin \text{Cone}(\mathcal{G}_4^{\text{el}}(S)),$$

then there is a function $q \in \mathbb{R}^S$ such that

$$q \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ, \quad \langle w, q \rangle > 0.$$

Every local exchange lies in a quadratic slice. In a quadratic slice, the only exchange with two distinct branches is the exchange between two disjoint off-diagonal pairs. If both branches were feasible, all six off-diagonal points would be present and the slice would be branching. Thus every local exchange has a unique feasible branch, and its inequality is represented by one of

(3.1), (3.2), or (3.3). It follows from the polar condition that q is M -concave. The family (2.2) lies in $\mathcal{L}_S^+$, whereas

$$R_w(f_t) = e^{t\langle w, q \rangle} \longrightarrow \infty.$$

This contradicts $w \in \text{BR}(\mathcal{L}_S^+)$. Therefore

$$w \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S)) \subseteq \text{Cone}(\mathcal{G}_4(S)).$$

Now suppose that some slice S_i is branching. By Lemma 3.3, either $S = e_i + T$ or $z_1, z_2, z_3, z_4 \in S$. In the first case, Lemma 3.4 gives

$$\text{BR}(\mathcal{L}_S^+) = \text{BR}(\mathcal{L}_T^+).$$

Since $S_i = T$, the natural identification of coefficient coordinates realizes w as the extension of an element of $\text{BR}(\mathcal{L}_{S_i}^+)$. Hence

$$w \in \mathcal{G}_4(S) \subseteq \text{Cone}(\mathcal{G}_4(S)).$$

It remains to treat the case $z_1, z_2, z_3, z_4 \in S$. Suppose that

$$w \notin \text{Cone}(\mathcal{G}_4^{\text{el}}(S)).$$

Choose

$$q \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ \quad \text{with} \quad \langle w, q \rangle > 0.$$

Let $\ell(\alpha) = \langle b, \alpha \rangle$ be the unique linear function such that $q' := q - \ell$ satisfies $q'(z_i) = 0$ for every i . By (3.4), $\langle h, \ell \rangle = 0$ for every $h \in \mathcal{G}_4^{\text{el}}(S)$, and therefore

$$q' \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ.$$

By (3.13), $\langle w, \ell \rangle = 0$, so $\langle w, q' \rangle > 0$.

Lemma 3.6 gives (3.6)–(3.9) for q' . By Lemma 3.7,

$$q' = \sum_{\rho} \lambda_{\rho} q^{(\rho)}, \quad \lambda_{\rho} \geq 0,$$

where every $q^{(\rho)}$ is M -concave. Since $\langle w, q' \rangle > 0$, some index ρ with $\lambda_{\rho} > 0$ satisfies

$$\langle w, q^{(\rho)} \rangle > 0.$$

Applying (2.2) to this $q^{(\rho)}$ produces a family in $\mathcal{L}_S^+$ along which R_w tends to infinity, again a contradiction. Thus

$$w \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S)) \subseteq \text{Cone}(\mathcal{G}_4(S)).$$

We have proved

$$\text{BR}(\mathcal{L}_S^+) \subseteq \text{Cone}(\mathcal{G}_4(S)).$$

Together with (3.5), this proves the theorem. $\square$

Remark 3.8 (Where the cubic hypothesis enters). The degree-three hypothesis enters in the branching analysis. Since

$$\Delta_4^{3-2} = \{e_1, e_2, e_3, e_4\},$$

every quadratic slice is indexed by a single coordinate. Lemma 3.3 then uses $|\alpha| = 3$ to reduce a branching support either to a translated quadratic support or to the case in which all four points $z_1, \dots, z_4$ are present. In the latter case, every point of Δ_4^3 is one of the z_i , a_{ij} , or p_i , which makes the support-graph decomposition possible.

4. CUT-CONE SEPARATION CERTIFICATES

The two minimal counterexamples will be separated from the quadratic local cone by functionals whose quadratic slices lie in cut cones. We first construct the basic polar directions associated with cuts and then turn them into a slice-by-slice certificate. The resulting criterion will certify both minimal obstructions and their subsequent propagations.

For a nontrivial cut $A \mid A^c$ of $[n]$, let δ_A be its cut semimetric:

$$\delta_A(i, j) = \begin{cases} 1, & \text{exactly one of } i, j \text{ lies in } A, \\ 0, & \text{otherwise.} \end{cases}$$

The cut cone is

$$\text{Cut}_n := \text{Cone}\{\delta_A : \emptyset \neq A \subsetneq [n]\}.$$

For $0 \leq s \leq d$, put

$$\mu_d(s) := \min\{s, d - s\}.$$

For $\alpha \in \Delta_n^d$, write

$$\alpha(A) := \sum_{i \in A} \alpha_i$$

and define

$$q_\alpha^A := \mu_d(\alpha(A)) = \min\{\alpha(A), d - \alpha(A)\}.$$

Lemma 4.1 (Cut directions). *For every nontrivial cut $A \mid A^c$ of $[n]$ and every $d \geq 1$,*

$$q^A \in \text{BR}(\mathcal{L}_{n,d}^+)^{\circ}.$$

Equivalently,

$$\langle w, q^A \rangle \leq 0 \quad \text{for every } w \in \text{BR}(\mathcal{L}_{n,d}^+).$$

Proof. When $d = 1$, the direction q^A is zero, so the assertion is immediate. Assume that $d \geq 2$ and consider

$$f_{A,t}(\mathbf{x}) := \sum_{\alpha \in \Delta_n^d} e^{t\mu_d(\alpha(A))} \frac{\mathbf{x}^\alpha}{\alpha!}, \quad t \geq 0.$$

We verify directly that $f_{A,t}$ is Lorentzian. Fix $\beta \in \Delta_n^{d-2}$ and put

$$s := \beta(A), \quad a := |\beta(A)|, \quad b := |\beta(A^c)|.$$

The Hessian $H_\beta(f_{A,t})$ is constant on each of its two diagonal blocks and on its off-diagonal blocks, with respective values

$$u := e^{t\mu_d(s+2)}, \quad v := e^{t\mu_d(s+1)}, \quad w := e^{t\mu_d(s)}.$$

The subspace of vectors having coordinate sum zero within each block lies in the kernel. On the orthonormal basis given by the normalized indicators of A and A^c , the remaining matrix is

$$\begin{pmatrix} au & \sqrt{ab}v \\ \sqrt{ab}v & bw \end{pmatrix}.$$

Its determinant is

$$ab(uw - v^2) \leq 0,$$

because the discrete concavity of μ_d gives

$$\mu_d(s+2) + \mu_d(s) \leq 2\mu_d(s+1).$$

Hence this 2×2 matrix, and therefore the full Hessian slice, has at most one positive eigenvalue. Since the full support Δ_n^d is M -convex, it follows that

$$f_{A,t} \in \mathcal{L}_{n,d}^+.$$

For $w' \in \text{BR}(\mathcal{L}_{n,d}^+)$,

$$R_{w'}(f_{A,t}) = e^{t\langle w', q^A \rangle}.$$

This expression must remain bounded as $t \rightarrow \infty$, which forces $\langle w', q^A \rangle \leq 0$. $\square$

We identify a symmetric array $Q = (Q_{ij})_{1 \leq i,j \leq n}$ with a vector on Δ_n^2 by setting

$$Q_{e_i+e_j} := Q_{ij}.$$

Thus

$$\langle v, Q \rangle = \sum_i v_{2e_i} Q_{ii} + \sum_{i < j} v_{e_i+e_j} Q_{ij},$$

so each off-diagonal coefficient is counted once. Define the *reduced symmetric array* of Q by

$$D_Q(i, j) := Q_{ij} - \frac{Q_{ii} + Q_{jj}}{2}.$$

The array D_Q has zero diagonal, but it need not be nonnegative or satisfy the triangle inequalities. When $D_Q \in \text{Cut}_n$, it is a cut semimetric.

Lemma 4.2 (Cut-cone certificate). *Let Q be a symmetric real array. If $D_Q \in \text{Cut}_n$, then*

$$\langle v, Q \rangle \leq 0 \quad \text{for every } v \in \text{BR}(\mathcal{L}_{n,2}^+).$$

Consequently, suppose that $q \in \mathbb{R}^{\Delta_n^d}$ has the property that, for every $\beta \in \Delta_n^{d-2}$, the reduced array

$$D_\beta^q(i, j) := q_{\beta+e_i+e_j} - \frac{q_{\beta+2e_i} + q_{\beta+2e_j}}{2}$$

belongs to Cut_n . Then

$$q \in \text{Cone}(\mathcal{G}_n(\Delta_n^d))^\circ.$$

Proof. First observe that every $v \in \text{BR}(\mathcal{L}_{n,2}^+)$ is balanced. Indeed, for $a \in \mathbb{R}^n$ and $t \in \mathbb{R}$, the diagonal change of variables $x_i \mapsto e^{ta_i} x_i$ preserves $\mathcal{L}_{n,2}^+$ and multiplies R_v by

$$\exp \left(t \left\langle a, \sum_{\gamma \in \Delta_n^2} v_\gamma \gamma \right\rangle \right).$$

Boundedness for both signs of t gives

$$\sum_{\gamma \in \Delta_n^2} v_\gamma \gamma = 0.$$

Consequently, v pairs trivially with every symmetric array of the form $L_{ij} = a_i + a_j$. More explicitly,

$$\langle v, L \rangle = \left\langle a, \sum_{\gamma \in \Delta_n^2} v_\gamma \gamma \right\rangle = 0.$$

Taking $a_i = Q_{ii}/2$, we have $Q = D_Q + L$, and hence

$$\langle v, Q \rangle = \langle v, D_Q \rangle.$$

For $d = 2$, the cut direction q^A from Lemma 4.1 is exactly the cut semimetric δ_A . If

$$D_Q = \sum_A \lambda_A \delta_A, \quad \lambda_A \geq 0,$$

then

$$\langle v, Q \rangle = \sum_A \lambda_A \langle v, \delta_A \rangle \leq 0.$$

For the final assertion, fix $\beta \in \Delta_n^{d-2}$ and set

$$Q_\beta(i, j) := q_{\beta+e_i+e_j}.$$

Its reduced array is D_β^q , which belongs to Cut_n by hypothesis. Therefore, for every $v \in \text{BR}(\mathcal{L}_{n,2}^+)$,

$$\langle \tilde{v}^\beta, q \rangle = \langle v, Q_\beta \rangle \leq 0.$$

Thus q is nonpositive on every quadratic local generator, and hence

$$q \in \text{Cone}(\mathcal{G}_n(\Delta_n^d))^\circ.$$

□

Lemma 4.3 (Reduced slices of a cut direction). *Fix a cut $A \mid A^c$ and $\beta \in \Delta_n^{d-2}$. Put*

$$s := \beta(A), \quad \lambda_s := \mu_d(s+1) - \frac{\mu_d(s) + \mu_d(s+2)}{2} \geq 0.$$

Then the cut direction q^A satisfies

$$D_\beta^{q^A} = \lambda_s \delta_A.$$

Proof. Let

$$\varepsilon_i := \begin{cases} 1, & i \in A, \\ 0, & i \notin A. \end{cases}$$

Then

$$q_{\beta+e_i+e_j}^A = \mu_d(s + \varepsilon_i + \varepsilon_j).$$

If i and j lie on the same side of the cut, the corresponding reduced entry is zero. If they lie on opposite sides, it equals

$$\mu_d(s+1) - \frac{\mu_d(s) + \mu_d(s+2)}{2} = \lambda_s.$$

The nonnegativity of λ_s follows from the discrete concavity of μ_d . □

5. THE TWO MINIMAL OBSTRUCTIONS

The failure region has exactly two minimal pairs, $(4, 4)$ and $(5, 3)$. In this section we construct a full-support bounded ratio outside the quadratic local cone at each pair. The two proofs share the same pattern: quadratic matrix inequalities prove boundedness, and a cut-cone certificate proves nonlocality.

5.1. The four-variable quartic obstruction. We begin with the minimal obstruction at $(n, d) = (4, 4)$. The bounded ratio comes from triangular quadratic inequalities and one four-point inequality, while a perturbed square-transportation direction separates it from the quadratic local cone.

For $\beta \in \Delta_4^2$ and pairwise distinct $i, j, k \in [4]$, set

$$T_\beta^{ij|k}(f) := \frac{c_{\beta+e_i+e_j} c_{\beta+2e_k}}{c_{\beta+e_i+e_k} c_{\beta+e_j+e_k}}.$$

Here and below, a digit string in a coefficient or slice subscript denotes the corresponding multi-index. The second inequality in Lemma 3.2 gives

$$T_\beta^{ij|k}(f) \leq 2.$$

Define

$$(5.1) \quad R_4(f) := \frac{c_{1030} c_{3010} c_{0202}}{c_{2020} c_{1012} c_{1210}},$$

and let w_4 denote its exponent vector. Place 1, 2, 3, 4 at the vertices of a square in cyclic order and let

$$P_1 := \{1, 4\} \mid \{2, 3\}, \quad P_2 := \{1, 2\} \mid \{3, 4\}.$$

For a cut $P = A \mid A^c$, write $q^P = q^A$ and $\delta_P = \delta_A$. Put

$$\alpha_* := (2, 0, 2, 0)$$

and define

$$(5.2) \quad q_4^0(\alpha) := \mu_4(\alpha_2 + \alpha_3) + \mu_4(\alpha_3 + \alpha_4) = q_\alpha^{P_1} + q_\alpha^{P_2}.$$

Finally, set

$$(5.3) \quad q_4(\alpha) := \begin{cases} q_4^0(\alpha_*) - 2, & \alpha = \alpha_*, \\ q_4^0(\alpha), & \alpha \neq \alpha_*. \end{cases}$$

Theorem 5.1 (Four-variable quartic obstruction). *For every $f \in \mathcal{L}_{4,4}^+$,*

$$R_4(f) \leq 64.$$

Moreover,

$$w_4 \in \text{BR}(\mathcal{L}_{4,4}^+) \setminus \text{Cone}(\mathcal{G}_4(\Delta_4^4)).$$

More precisely, every reduced quadratic slice of q_4 belongs to Cut_4 , and

$$\langle w_4, q_4 \rangle = 2 > 0, \quad q_4 \in \text{Cone}(\mathcal{G}_4(\Delta_4^4))^\circ.$$

Proof. Fix $\beta_0 = (0, 1, 0, 1)$ and write $H_{\beta_0}(f) = (p_{ij})$. Set

$$A := c_{1201} c_{0112} = p_{12} p_{34}, \quad B := c_{1111} c_{0202} = p_{13} p_{24}, \quad C := c_{1102} c_{0211} = p_{14} p_{23}.$$

The four-point part of Lemma 3.2 gives

$$\sqrt{B} \leq \sqrt{A} + \sqrt{C}.$$

Consequently,

$$(5.4) \quad \min \left\{ \frac{B}{A}, \frac{B}{C} \right\} \leq 4.$$

Indeed, if both ratios were greater than 4, then $\sqrt{A} < \sqrt{B}/2$ and $\sqrt{C} < \sqrt{B}/2$, contradicting the preceding triangle inequality.

For brevity, write $R_4 = R_4(f)$. Direct cancellation gives

$$(5.5) \quad R_4 = T_{1010}^{24|3} T_{1010}^{43|1} T_{1100}^{24|3} T_{0011}^{24|1} \frac{B}{A},$$

$$(5.6) \quad R_4 = T_{1010}^{24|3} T_{1010}^{23|1} T_{0110}^{24|1} T_{1001}^{24|3} \frac{B}{C}.$$

Choose the factorization corresponding to the smaller ratio in (5.4). Since each of its four triangular factors is at most 2, we obtain

$$R_4(f) \leq 2^4 \cdot 4 = 64.$$

It remains to prove that the bounded ratio is not generated by the quadratic slices. The values of q_4^0 on the three numerator coordinates of (5.1) are

$$2, \quad 2, \quad 4,$$

whereas its values on the three denominator coordinates are

$$4, \quad 2, \quad 2.$$

Hence $\langle w_4, q_4^0 \rangle = 0$. Since $\alpha_* = (2, 0, 2, 0)$ occurs in the denominator of R_4 , (5.3) gives

$$(5.7) \quad \langle w_4, q_4 \rangle = 2 > 0.$$

By Lemma 4.3, every reduced slice of $q_4^0 = q^{P_1} + q^{P_2}$ belongs to Cut_4 . The modification at α_* can affect only the three slices

$$\beta_1 := \alpha_* - 2e_1 = (0, 0, 2, 0), \quad \beta_{13} := \alpha_* - e_1 - e_3 = (1, 0, 1, 0), \quad \beta_3 := \alpha_* - 2e_3 = (2, 0, 0, 0).$$

In the β_1 -slice, lowering the diagonal entry Q_{11} by 2 adds $\delta_{\{1\}}$ to the reduced array. Likewise, in the β_3 -slice it adds $\delta_{\{3\}}$. Thus both modified arrays remain in Cut_4 .

For the middle slice, Lemma 4.3 gives coefficient

$$\mu_4(2) - \frac{\mu_4(1) + \mu_4(3)}{2} = 1$$

for both cuts. Before the modification, its reduced semimetric is $\delta_{P_1} + \delta_{P_2}$: it has value 1 on the four edges of the square and value 2 on the two diagonals. Here α_* is the $(1, 3)$ off-diagonal entry. Lowering it by 2 changes the reduced value on $\{1, 3\}$ from 2 to zero, and the resulting semimetric is

$$\delta_{\{2\}} + \delta_{\{4\}} \in \text{Cut}_4.$$

Therefore every reduced quadratic slice of q_4 belongs to Cut_4 . Lemma 4.2 yields

$$(5.8) \quad q_4 \in \text{Cone}(\mathcal{G}_4(\Delta_4^4))^\circ.$$

Together with (5.7), this proves

$$w_4 \notin \text{Cone}(\mathcal{G}_4(\Delta_4^4)).$$

The bound already proved gives $w_4 \in \text{BR}(\mathcal{L}_{4,4}^+)$, and hence the claimed strict inclusion follows from (2.1). $\square$

5.2. The five-variable cubic obstruction. We next treat the other minimal failure pair, $(n, d) = (5, 3)$.

Theorem 5.2 (Five-variable cubic obstruction). *Let $S = \Delta_5^3$. Writing $c_{(a,b,c,d,e)} = c_{abcde}$, define*

$$(5.9) \quad R_5(f) := \frac{c_{30000}c_{21000}c_{10101}c_{00210}c_{00021}}{c_{20100}c_{20010}c_{20001}c_{01110}c_{00111}}.$$

Then

$$R_5(f) \leq 32 \quad \text{for every } f \in \mathcal{L}_{5,3}^+,$$

but the exponent vector of R_5 does not belong to $\text{Cone}(\mathcal{G}_5(\Delta_5^3))$. In particular,

$$\text{Cone}(\mathcal{G}_5(\Delta_5^3)) \subsetneq \text{BR}(\mathcal{L}_{5,3}^+).$$

The proof has two parts. The upper bound is obtained by combining three triangular quadratic inequalities with one four-point inequality. The failure of local generation is then detected by perturbing a sum of cut directions while keeping every reduced quadratic slice inside the cut cone.

Proof. Let w_R be the exponent vector of R_5 , and write

$$a_{ijk} := c_{e_i+e_j+e_k},$$

where indices may repeat and their order is immaterial. For $m \in [5]$ and pairwise distinct $i, j, k \in [5]$, set

$$T_m^{ij|k} := \frac{a_{mij}a_{mkk}}{a_{mik}a_{mjk}}.$$

Here m is unrestricted and may coincide with one of i, j, k . By the second inequality in Lemma 3.2,

$$T_m^{ij|k} \leq 2.$$

Set

$$A := a_{112}a_{134}, \quad B := a_{113}a_{124}, \quad C := a_{114}a_{123}.$$

Applying the four-point inequality from Lemma 3.2 to the slice $H_{e_1}(f)$ gives

$$\sqrt{A} \leq \sqrt{B} + \sqrt{C}.$$

Consequently,

$$(5.10) \quad \min \left\{ \frac{A}{B}, \frac{A}{C} \right\} \leq 4.$$

Indeed, if both ratios were greater than 4, then $\sqrt{B} < \sqrt{A}/2$ and $\sqrt{C} < \sqrt{A}/2$, contradicting the preceding triangle inequality.

For brevity, write $R_5 = R_5(f)$. Direct cancellation gives the two factorizations

$$(5.11) \quad R_5 = T_1^{45|1} T_4^{12|3} T_5^{13|4} \frac{A}{B},$$

$$(5.12) \quad R_5 = T_1^{35|1} T_3^{12|4} T_4^{45|3} \frac{A}{C}.$$

Choose the factorization corresponding to the smaller ratio in (5.10). Since each of its three T -factors is at most 2, we obtain

$$R_5(f) \leq 2^3 \cdot 4 = 32.$$

Thus

$$w_R \in \text{BR}(\mathcal{L}_{5,3}^+).$$

We now separate w_R from the quadratic local cone using Lemma 4.2. Put

$$\mu(s) := \min\{s, 3 - s\}$$

and define

$$q_\alpha^0 := \sum_{r=3}^5 \mu(\alpha_2 + \alpha_r), \quad \rho := \delta_{\{2,3\}} + \delta_{\{2,4\}} + \delta_{\{2,5\}}.$$

In the notation of Lemma 4.1,

$$q^0 = q^{\{2,3\}} + q^{\{2,4\}} + q^{\{2,5\}}.$$

For every $m, i, j \in [5]$,

$$q_{e_m+e_i+e_j}^0 = \frac{\rho(m, i) + \rho(m, j) + \rho(i, j)}{2}.$$

Thus, if

$$D_m^0(i, j) := q_{e_m+e_i+e_j}^0 - \frac{q_{e_m+2e_i}^0 + q_{e_m+2e_j}^0}{2},$$

then

$$2D_m^0 = \rho \quad \text{for every } m \in [5].$$

We now perturb a single coordinate. Set

$$q_{00111} := q_{00111}^0 - 1 = 2,$$

and let $q_\alpha := q_\alpha^0$ for every other $\alpha \in \Delta_5^3$. If D_m denotes the reduced array of q in the m th quadratic slice, then a direct calculation gives

$$\begin{aligned} 2D_1 &= 2D_2 = \delta_{\{2,3\}} + \delta_{\{2,4\}} + \delta_{\{2,5\}}, \\ 2D_3 &= \delta_{\{2,3\}} + \delta_{\{2\}} + \delta_{\{2,4,5\}}, \\ 2D_4 &= \delta_{\{2,4\}} + \delta_{\{2\}} + \delta_{\{2,3,5\}}, \\ 2D_5 &= \delta_{\{2,5\}} + \delta_{\{2\}} + \delta_{\{2,3,4\}}. \end{aligned}$$

Each D_m therefore belongs to Cut_5 . These are all the quadratic slices, since

$$\Delta_5^{3-2} = \{e_1, e_2, e_3, e_4, e_5\}.$$

Lemma 4.2 now gives

$$q \in \text{Cone}(\mathcal{G}_5(\Delta_5^3))^\circ.$$

Finally, the q^0 -values on the five numerator coordinates of (5.9) are

$$0, 3, 2, 2, 2,$$

whereas those on the five denominator coordinates are

$$1, 1, 1, 3, 3.$$

Both lists sum to 9, so

$$\langle w_R, q^0 \rangle = 0.$$

Since c_{00111} occurs in the denominator of R_5 and its q -coordinate was decreased by 1,

$$\langle w_R, q \rangle = 1 > 0.$$

Because q lies in the polar of the quadratic local cone, this proves

$$w_R \notin \text{Cone}(\mathcal{G}_5(\Delta_5^3)).$$

Together with the upper bound and (2.1), the strict inclusion follows. $\square$

6. PROPAGATION AND COMPLETION OF THE CLASSIFICATION

We now propagate the two minimal obstructions. First we keep four variables and extend the quartic obstruction through every higher degree. We then fix the degree and use variable aggregation to pass from each minimal number of variables to every larger one. The resulting two failure regions, together with the positive results of Sections 2 and 3, complete the classification.

6.1. Degree propagation in four variables. The quartic bounded ratio extends to higher degree by differentiation. To preserve the separation from the quadratic local cone, we construct a degree-uniform family of perturbed square-transportation directions.

Fix $d \geq 4$ and write

$$m := d - 4, \quad a := \left\lfloor \frac{m}{2} \right\rfloor, \quad b := \left\lceil \frac{m}{2} \right\rceil, \quad \gamma := (a, 0, b, 0).$$

Notice that m and d have the same parity. Define

$$(6.1) \quad R_d(f) := \frac{c_{\gamma+(1,0,3,0)} c_{\gamma+(3,0,1,0)} c_{\gamma+(0,2,0,2)}}{c_{\gamma+(2,0,2,0)} c_{\gamma+(1,0,1,2)} c_{\gamma+(1,2,1,0)}}.$$

In factorial normalization, differentiation by ∂^γ preserves the displayed coefficients. Hence

$$(6.2) \quad R_d(f) = R_d(\partial^\gamma f).$$

Let w_d denote the exponent vector of R_d , and put

$$\alpha_* := \gamma + (2, 0, 2, 0), \quad \varepsilon_d := \begin{cases} 2, & d \text{ even}, \\ 1, & d \text{ odd}. \end{cases}$$

Place 1, 2, 3, 4 at the vertices of a square in cyclic order and let

$$P_1 := \{1, 4\} \mid \{2, 3\}, \quad P_2 := \{1, 2\} \mid \{3, 4\}.$$

For a cut $P = A \mid A^c$, write $q^P = q^A$ and $\delta_P = \delta_A$; both are independent of the chosen side. Using the notation μ_d from Section 4, define

$$(6.3) \quad q_d^0(\alpha) := \mu_d(\alpha_2 + \alpha_3) + \mu_d(\alpha_3 + \alpha_4) = q_\alpha^{P_1} + q_\alpha^{P_2}.$$

Finally, set

$$(6.4) \quad q_d(\alpha) := \begin{cases} q_d^0(\alpha_*) - \varepsilon_d, & \alpha = \alpha_*, \\ q_d^0(\alpha), & \alpha \neq \alpha_*. \end{cases}$$

Theorem 6.1 (Degree propagation in four variables). *For every $d \geq 4$,*

$$w_d \in \text{BR}(\mathcal{L}_{4,d}^+) \setminus \text{Cone}(\mathcal{G}_4(\Delta_4^d)).$$

More precisely,

$$R_d(f) \leq 64 \quad \text{for every } f \in \mathcal{L}_{4,d}^+,$$

and

$$\langle w_d, q_d \rangle = \varepsilon_d > 0, \quad q_d \in \text{Cone}(\mathcal{G}_4(\Delta_4^d))^\circ.$$

Proof. The case $d = 4$ is Theorem 5.1. The following argument, which also recovers that case, treats every $d \geq 4$ uniformly. If $f \in \mathcal{L}_{4,d}^+$, then $\partial^\gamma f$ is a full-support Lorentzian quartic. By (6.2) and Theorem 5.1,

$$R_d(f) = R_d(\partial^\gamma f) \leq 64.$$

Therefore

$$w_d \in \text{BR}(\mathcal{L}_{4,d}^+).$$

We next compute the pairing with the square-transportation direction. If m is even, the values of q_d^0 on the three numerator coordinates in (6.1) are

$$m + 2, \quad m + 2, \quad m + 4,$$

whereas its values on the three denominator coordinates are

$$m + 4, \quad m + 2, \quad m + 2.$$

If m is odd, the corresponding numerator values are

$$m + 1, \quad m + 3, \quad m + 3,$$

and the denominator values are

$$m + 3, \quad m + 2, \quad m + 2.$$

In either parity, the numerator and denominator values have the same sum. Hence

$$\langle w_d, q_d^0 \rangle = 0.$$

Since α_* occurs in the denominator of R_d , (6.4) gives

$$(6.5) \quad \langle w_d, q_d \rangle = \varepsilon_d > 0.$$

It remains to prove that q_d is nonpositive on the quadratic local cone. By Lemma 4.3, every reduced slice of $q_d^0 = q^{P_1} + q^{P_2}$ belongs to Cut_4 .

The modification at α_* can affect a slice indexed by β only if

$$\alpha_* = \beta + e_i + e_j$$

for some $i, j \in [4]$. Since α_* is supported only on coordinates 1 and 3, one must have $i, j \in \{1, 3\}$. Thus precisely the following three slices can be affected:

$$\beta_1 := \alpha_* - 2e_1, \quad \beta_{13} := \alpha_* - e_1 - e_3, \quad \beta_3 := \alpha_* - 2e_3.$$

In the β_1 -slice, the coefficient at α_* is the diagonal entry Q_{11} . Lowering it by ε_d adds $(\varepsilon_d/2)\delta_{\{1\}}$ to the reduced array. Likewise, in the β_3 -slice it adds $(\varepsilon_d/2)\delta_{\{3\}}$. Since the original reduced arrays belong to Cut_4 , both modified arrays remain in Cut_4 .

It remains to consider the middle slice

$$\beta_{13} = \gamma + e_1 + e_3 = (a + 1, 0, b + 1, 0).$$

For the chosen sides $\{2, 3\}$ of P_1 and $\{3, 4\}$ of P_2 , the cut mass of β_{13} is $b + 1$. Lemma 4.3 therefore gives the same coefficient for both cuts, namely

$$\lambda_d := \mu_d(b + 2) - \frac{\mu_d(b + 1) + \mu_d(b + 3)}{2} = \begin{cases} 1, & d \text{ even}, \\ \frac{1}{2}, & d \text{ odd} \end{cases} = \frac{\varepsilon_d}{2}.$$

Before the modification, the reduced semimetric in this slice is

$$\lambda_d(\delta_{P_1} + \delta_{P_2}).$$

It has value λ_d on the four edges of the square and $2\lambda_d$ on its two diagonals. The coefficient α_* is the $(1, 3)$ off-diagonal entry of this slice. Lowering it by $\varepsilon_d = 2\lambda_d$ changes only the reduced value on $\{1, 3\}$, from $2\lambda_d$ to zero. The resulting semimetric is

$$\lambda_d(\delta_{\{2\}} + \delta_{\{4\}}) \in \text{Cut}_4.$$

We have shown that every reduced quadratic slice of q_d belongs to Cut_4 . Lemma 4.2 therefore yields

$$(6.6) \quad q_d \in \text{Cone}(\mathcal{G}_4(\Delta_4^d))^\circ.$$

Combining (6.5) and (6.6), we obtain

$$w_d \notin \text{Cone}(\mathcal{G}_4(\Delta_4^d)).$$

Together with $w_d \in \text{BR}(\mathcal{L}_{4,d}^+)$ and (2.1), this also gives

$$\text{Cone}(\mathcal{G}_4(\Delta_4^d)) \subsetneq \text{BR}(\mathcal{L}_{4,d}^+).$$

□

6.2. Dimension propagation by variable aggregation. We now propagate a full-support obstruction equipped with a cut-cone certificate to arbitrarily many variables.

Lemma 6.2 (Variable aggregation). *Let $m \geq 2$ and $d \geq 2$. Suppose that $w \in \text{BR}(\mathcal{L}_{m,d}^+)$ and $q^{(0)} \in \mathbb{R}^{\Delta_m^d}$ satisfy*

$$\langle w, q^{(0)} \rangle > 0, \quad D_\beta^{q^{(0)}} \in \text{Cut}_m \quad \text{for every } \beta \in \Delta_m^{d-2}.$$

Then, for every $n \geq m$, there exist

$$\tilde{w} \in \text{BR}(\mathcal{L}_{n,d}^+) \quad \text{and} \quad q \in \text{Cone}(\mathcal{G}_n(\Delta_n^d))^\circ$$

such that

$$\langle \tilde{w}, q \rangle > 0.$$

In particular,

$$\tilde{w} \notin \text{Cone}(\mathcal{G}_n(\Delta_n^d)).$$

Proof. Choose a surjection $\pi : [n] \rightarrow [m]$ such that $\pi(i) = i$ for $1 \leq i \leq m$, and define the aggregation map

$$A_\pi : \Delta_n^d \longrightarrow \Delta_m^d, \quad (A_\pi \alpha)_r := \sum_{\pi(i)=r} \alpha_i.$$

Extend w by zero to the coordinates of Δ_n^d that are not supported on the first m variables; denote the resulting vector by $\tilde{w}$. If $f \in \mathcal{L}_{n,d}^+$, then

$$g(x_1, \dots, x_m) := f(x_1, \dots, x_m, 0, \dots, 0)$$

has full support Δ_m^d . Moreover, every quadratic Hessian slice of g is a principal submatrix of the corresponding slice of f . Hence $g \in \mathcal{L}_{m,d}^+$, and

$$R_{\tilde{w}}(f) = R_w(g).$$

It follows that

$$\tilde{w} \in \text{BR}(\mathcal{L}_{n,d}^+).$$

Now define

$$q_\alpha := q_{A_\pi \alpha}^{(0)} \quad (\alpha \in \Delta_n^d).$$

For $\beta \in \Delta_n^{d-2}$, a direct calculation gives

$$D_\beta^q(i, j) = D_{A_\pi \beta}^{q^{(0)}}(\pi(i), \pi(j)).$$

Thus D_β^q is the pullback under π of an element of Cut_m . Indeed, the pullback of a cut semimetric δ_A is

$$\delta_{\pi^{-1}(A)}.$$

Because π is surjective, $\pi^{-1}(A)$ is a nontrivial cut of $[n]$ whenever A is a nontrivial cut of $[m]$. It follows that

$$D_\beta^q \in \text{Cut}_n.$$

Lemma 4.2 therefore gives

$$q \in \text{Cone}(\mathcal{G}_n(\Delta_n^d))^\circ.$$

Finally, since π restricts to the identity on $[m]$,

$$\langle \tilde{w}, q \rangle = \langle w, q^{(0)} \rangle > 0.$$

This proves the result. $\square$

Corollary 6.3 (Sharp dimension boundary for cubics). *For every $n \geq 5$,*

$$\text{Cone}(\mathcal{G}_n(\Delta_n^3)) \subsetneq \text{BR}(\mathcal{L}_{n,3}^+).$$

Thus, for cubics, the quadratic local-to-global principle holds for every M -convex support if and only if $n \leq 4$.

Proof. The proof of Theorem 5.2 constructs a separator q whose reduced array in every quadratic slice is explicitly a nonnegative combination of cut semimetrics. Hence the hypotheses of Lemma 6.2 hold with $m = 5$ and $d = 3$. Its conclusion gives the strict inclusion for every $n \geq 5$. The positive statement for $n \leq 3$ follows from Proposition 2.2, and the case $n = 4$ is Theorem 3.1. $\square$

Corollary 6.4 (Full-support failure for all $n \geq 4$, $d \geq 4$). *For every $n \geq 4$ and $d \geq 4$,*

$$\text{Cone}(\mathcal{G}_n(\Delta_n^d)) \subsetneq \text{BR}(\mathcal{L}_{n,d}^+).$$

Proof. For $n = 4$, this is Theorem 6.1. The same theorem provides a separator q_d whose every reduced quadratic slice belongs to Cut_4 . Lemma 6.2 therefore lifts the obstruction to every $n \geq 4$. $\square$

6.3. Proof of the complete classification. We now combine the positive results and the two families of obstructions to prove the main theorem.

Proof of Theorem 1.1. Assume first that

$$n \leq 3, \quad \text{or} \quad d = 2, \quad \text{or} \quad (n, d) = (4, 3).$$

Proposition 2.2 proves the first two cases, while Theorem 3.1 proves the exceptional quaternary cubic case. Hence

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}_n(S))$$

for every nonempty M -convex support $S \subseteq \Delta_n^d$.

Conversely, suppose that none of the three conditions holds. Since $d \geq 2$, there are only two possibilities. If $d = 3$, then necessarily $n \geq 5$, and Corollary 6.3 gives a strict full-support counterexample. If $d \geq 4$, then necessarily $n \geq 4$, and Corollary 6.4 gives a strict full-support counterexample. In either case, Δ_n^d is M -convex, so the full-support counterexample disproves the universal equality in (i). Thus universal quadratic local-to-global generation fails already for $S = \Delta_n^d$. This proves both the equivalence and the final assertion. $\square$

The constants 32 and 64 are used only to establish boundedness; no claim of optimality is made.

The classification may be summarized by the table

	$d = 2$	$d = 3$	$d \geq 4$
$n \leq 3$	✓	✓	✓
$n = 4$	✓	✓	×
$n \geq 5$	✓	×	×

where ✓ means equality for every M -convex support, and × means failure already for full support.

7. A MEDIAN-GRAPH INTERPRETATION OF TRANSPORTATION POLAR DIRECTIONS

We conclude with a conceptual interpretation of the transportation costs underlying the separating functionals above. Transportation costs on finite median graphs give polar directions for full-support bounded-ratio cones in arbitrary dimension and degree.

Throughout this section, let $n \geq 2$ and $d \geq 1$. When $d = 1$, $\mathcal{L}_{n,1}^+$ denotes the linear forms with all coefficients positive; all transportation directions considered below are zero.

We first recall the standard cut structure of finite median graphs [3]. A connected graph is *median* if every triple of vertices has a unique vertex lying on a shortest path between each pair. Let G be a finite median graph with vertex set $[n]$. Its edge Θ -classes determine convex cuts

$$P_r = H_r \mid H_r^c, \quad 1 \leq r \leq m,$$

such that

$$(7.1) \quad d_G(u, v) = \#\{r : P_r \text{ separates } u \text{ and } v\}.$$

We shall also use the Helly property for convex subsets of a finite median graph. For completeness, suppose that $C_1, \dots, C_k$ is a minimal pairwise-intersecting family of convex sets with empty total intersection. Then $k \geq 3$, and minimality allows us to choose

$$x_i \in \bigcap_{j \neq i} C_j \quad (1 \leq i \leq k).$$

Let x be the median of x_1, x_2, x_3 . Every C_j contains at least two of these three vertices. Since C_j is convex and x lies on a shortest path between each pair, we have $x \in C_j$ for every j . This contradicts the assumption that the total intersection is empty.

For $\alpha \in \Delta_n^d$, define the transportation cost

$$(7.2) \quad q_\alpha^G := \min_{v \in [n]} \sum_{i=1}^n \alpha_i d_G(i, v).$$

Thus q_α^G is the minimum cost of moving the d tokens described by α to a common vertex. For a cut $P = H \mid H^c$, write

$$q^P := q^H = q^{H^c},$$

using the cut directions introduced in Section 4.

Theorem 7.1 (Median-graph transportation directions). *Let G be a finite median graph on $[n]$. Then*

$$q^G = \sum_{r=1}^m q^{P_r},$$

and consequently

$$q^G \in \text{BR}(\mathcal{L}_{n,d}^+)^{\circ}.$$

Equivalently, every $w \in \text{BR}(\mathcal{L}_{n,d}^+)$ satisfies

$$\langle w, q^G \rangle \leq 0.$$

Proof. Fix $\alpha \in \Delta_n^d$. For $v \in [n]$, let $H_r(v)$ denote the side of P_r that does not contain v . By (7.1), we may interchange the sums over tokens and cuts to obtain

$$\sum_{i=1}^n \alpha_i d_G(i, v) = \sum_{r=1}^m \alpha(H_r(v)) \geq \sum_{r=1}^m \min\{\alpha(H_r), \alpha(H_r^c)\}.$$

This gives a lower bound independent of v .

For each r with $\alpha(H_r) \neq \alpha(H_r^c)$, let M_r be the side of P_r having mass greater than $d/2$. These majority sides are pairwise intersecting. Indeed, if $M_r \cap M_s = \emptyset$, then

$$d \geq \alpha(M_r) + \alpha(M_s) > d,$$

a contradiction. Each M_r is convex, so the Helly property gives

$$\bigcap_r M_r \neq \emptyset.$$

If there are no cuts with unequal masses, choose any vertex $v \in [n]$. Otherwise, choose $v \in \bigcap_r M_r$.

For every cut with unequal masses, the chosen vertex v lies on the majority side, so the tokens crossing that cut have exactly the minority mass. For a cut with equal masses, either side gives the same contribution. Hence the preceding lower bound is attained, and

$$q_\alpha^G = \sum_{r=1}^m \min\{\alpha(H_r), \alpha(H_r^c)\} = \sum_{r=1}^m q_\alpha^{P_r}.$$

Since this holds for every $\alpha \in \Delta_n^d$,

$$q^G = \sum_{r=1}^m q^{P_r}.$$

By Lemma 4.1, each q^{P_r} belongs to $\text{BR}(\mathcal{L}_{n,d}^+)^{\circ}$. This polar is a convex cone, so their sum also belongs to it. Therefore

$$q^G \in \text{BR}(\mathcal{L}_{n,d}^+)^{\circ}.$$

□

Remark 7.2. For the square $G = C_4$ with the labeling used in Section 6.1, the two Θ -classes are exactly the two coordinate cuts used in that section. Thus Theorem 7.1 shows that the unmodified square transportation direction is nonpositive on the entire bounded-ratio cone $\text{BR}(\mathcal{L}_{4,d}^+)$ for every $d \geq 1$. The obstruction in Section 6.1 is obtained by modifying this global polar direction at a single coefficient while keeping every reduced quadratic slice in the cut cone.

More generally, taking G to be the hypercube Q_k gives the Hamming transportation direction in 2^k variables and every degree.

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DEPARTMENT OF MATHEMATICS, UNIVERSITY OF WISCONSIN–MADISON, MADISON, WI, USA
Email address: `dchen426@wisc.edu`

INSTITUTE OF MATHEMATICAL SCIENCES, SHANGHAITECH UNIVERSITY, SHANGHAI, CHINA
Email address: `ganbw2023@shanghaitech.edu.cn`

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF WISCONSIN–MADISON, MADISON, WI, USA
Email address: `ivy.liu@wisc.edu`

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF WISCONSIN–MADISON, MADISON, WI, USA
Email address: `zwang3694@wisc.edu`

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF WISCONSIN–MADISON, MADISON, WI, USA
Email address: `cwu493@wisc.edu`